

Customer Shopping Behavior Analysis

Analyzing 3,900 purchases to uncover insights into spending patterns, customer segments, and product preferences that drive strategic business decisions.



Dataset Overview

3,900

Total Purchases

Transactions analyzed
across various product
categories

18

Data Features

Columns capturing
demographics, purchases,
and behavior

50

Locations

Geographic coverage across
customer base

25

Products

Unique items purchased by
customers

Customer Data

- Age, Gender, Location
- Subscription Status
- Purchase History

Transaction Details

- Item, Category, Amount
- Season, Size, Color
- Discounts & Promotions



Data Preparation in Python

01

Data Loading & Exploration

Imported dataset using pandas and examined structure with `df.info()` and `.describe()` for summary statistics

02

Missing Data Handling

Imputed 37 missing Review Rating values using median rating of each product category

03

Feature Engineering

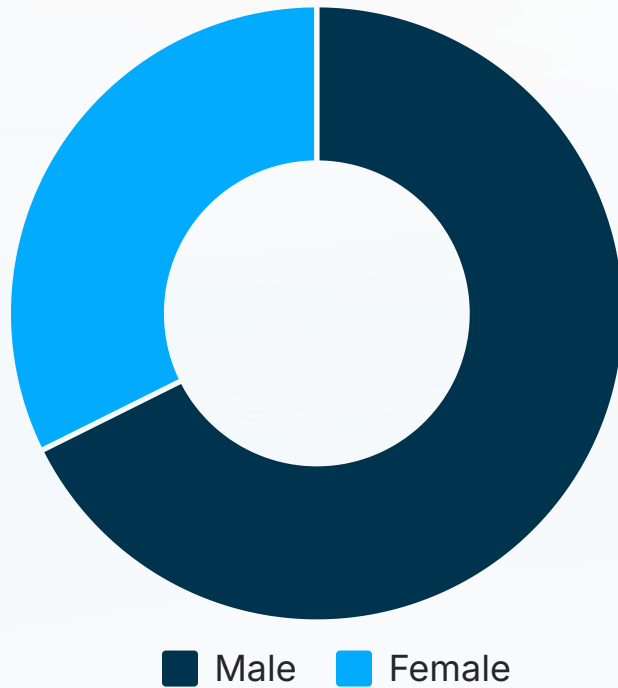
Created `age_group` bins and `purchase_frequency_days` columns for deeper analysis

04

Database Integration

Connected to PostgreSQL and loaded cleaned DataFrame for SQL analysis

Revenue Insights by Gender



Male Customers Drive Revenue

Male customers generated **\$157,890** in total revenue, more than double the **\$75,191** from female customers.

This significant gap suggests opportunities for targeted marketing to female segments.

High-Value Discount Users



839 Customers

Used discounts but still spent above average purchase amount



Smart Spenders

These customers maximize value while maintaining high purchase amounts



Key Segment

Prime candidates for loyalty programs and exclusive offers



Top-Rated Products & Shipping Analysis

Highest-Rated Products

- **Gloves**
3.86 average rating
- **Sandals**
3.84 average rating
- **Boots**
3.82 average rating
- **Hat**
3.80 average rating
- **Skirt**
3.78 average rating

Shipping Type Comparison

Express Shipping

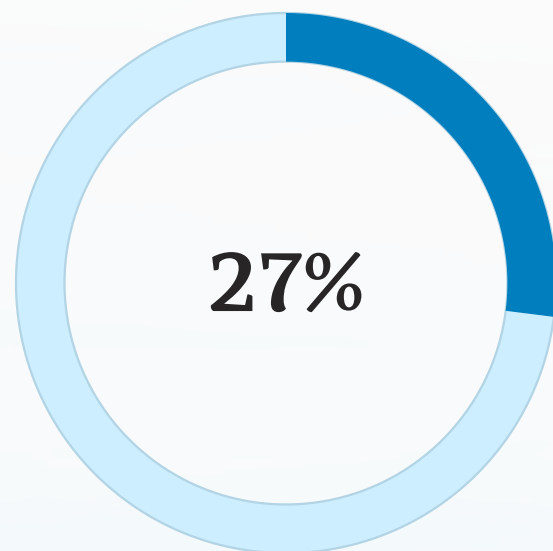
\$60.48 average purchase

Standard Shipping

\$58.46 average purchase

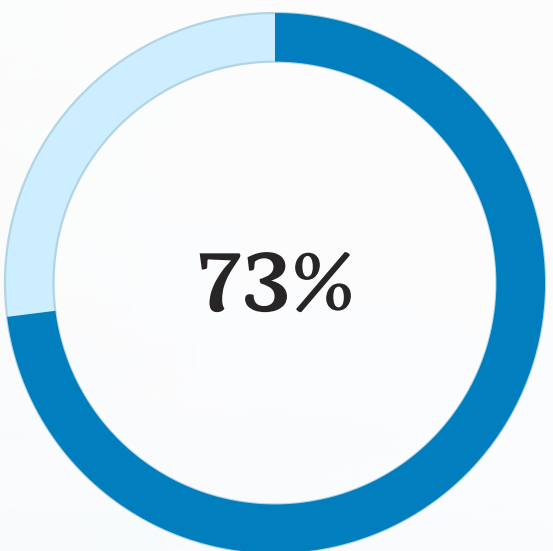
Express shipping customers spend slightly more per transaction, indicating higher purchase intent.

Subscription Status Analysis



Subscribers

1,053 customers with subscription status



Non-Subscribers

2,847 customers without subscriptions

Subscribers

Average spend: **\$59.49**

Total revenue: **\$62,645**

Non-Subscribers

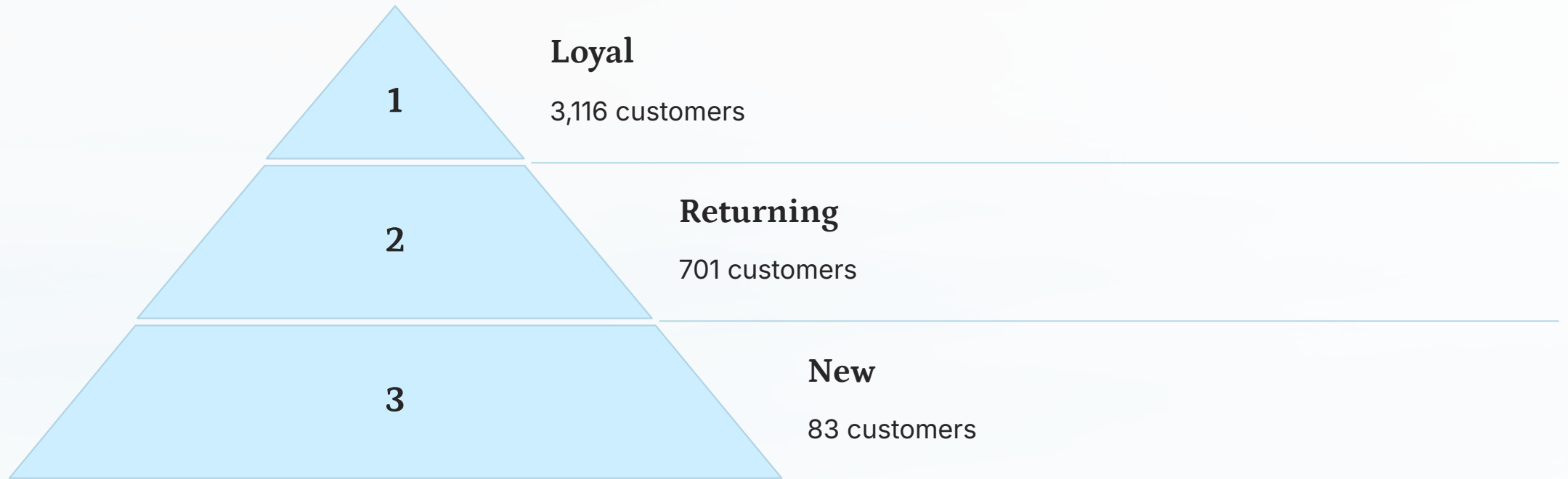
Average spend: **\$59.87**

Total revenue: **\$170,436**

Key Insight

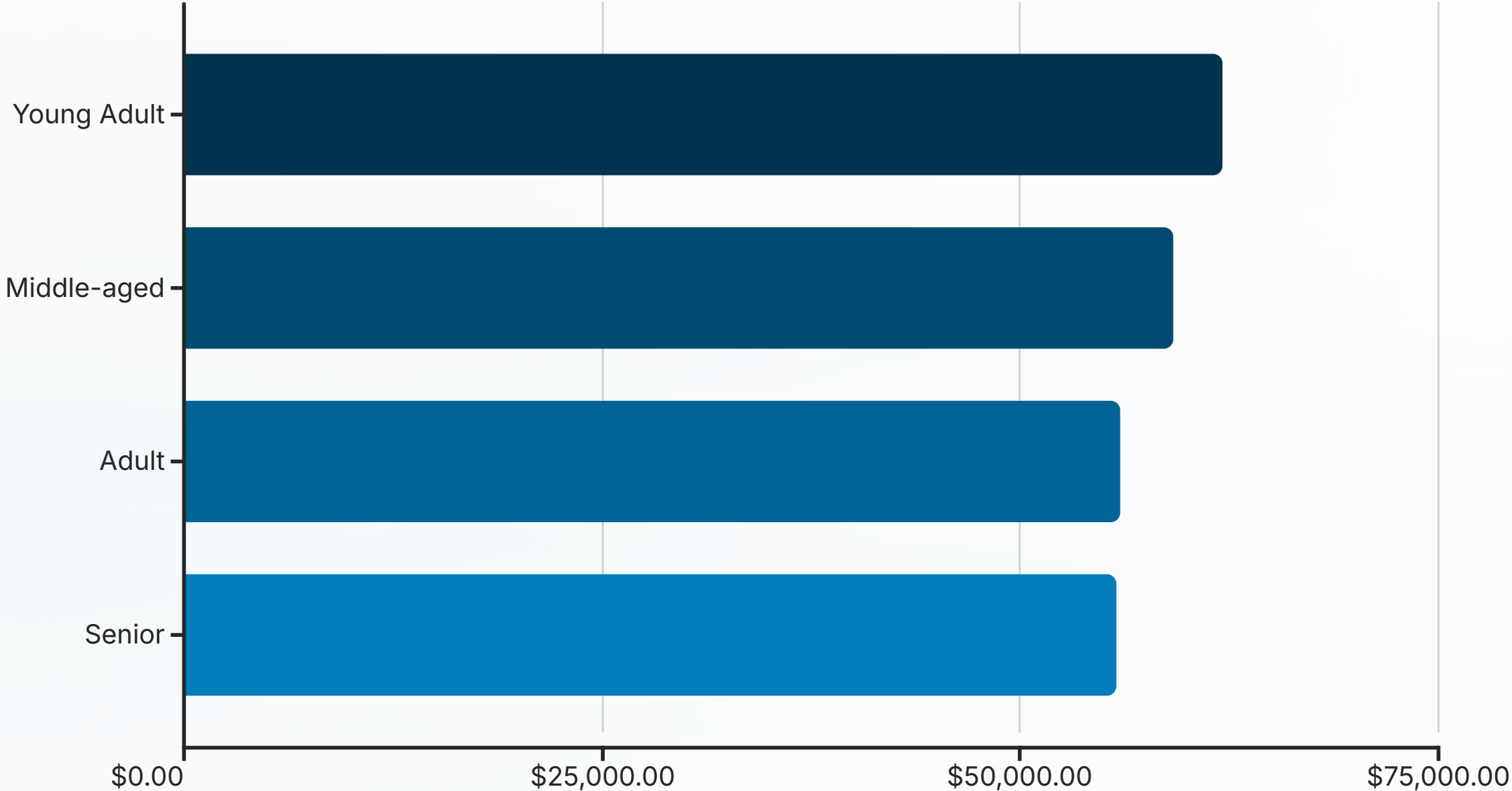
Similar spending patterns suggest untapped potential in converting non-subscribers.

Customer Segmentation



The majority of customers (80%) fall into the Loyal segment, demonstrating strong retention. However, only 83 new customers suggest opportunities to improve acquisition strategies.

Revenue by Age Group



Young Adults lead revenue generation at **\$62,143**, followed closely by Middle-aged customers at **\$59,197**. Revenue distribution is relatively balanced across age groups.

Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base with similar spending patterns.

Customer Loyalty Programs

Reward repeat buyers to strengthen the 3,116-customer Loyal segment and move Returning customers upward.

Review Discount Policy

Balance sales boosts with margin control—839 high-spenders use discounts effectively.

Product Positioning

Highlight top-rated products (Gloves, Sandals, Boots) in marketing campaigns to drive conversions.

Targeted Marketing

Focus on Young Adults and express-shipping users who demonstrate higher purchase intent and revenue contribution.